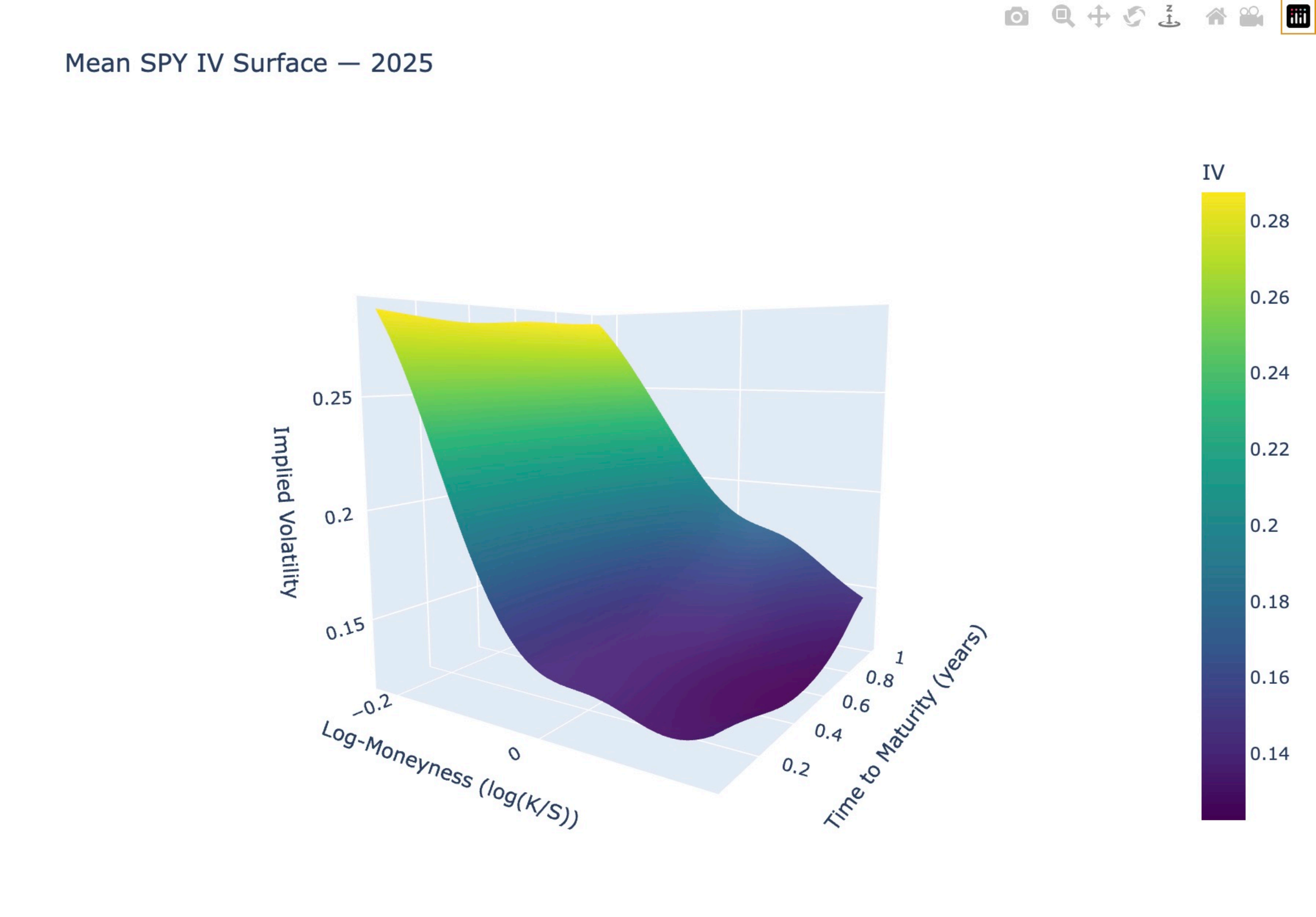
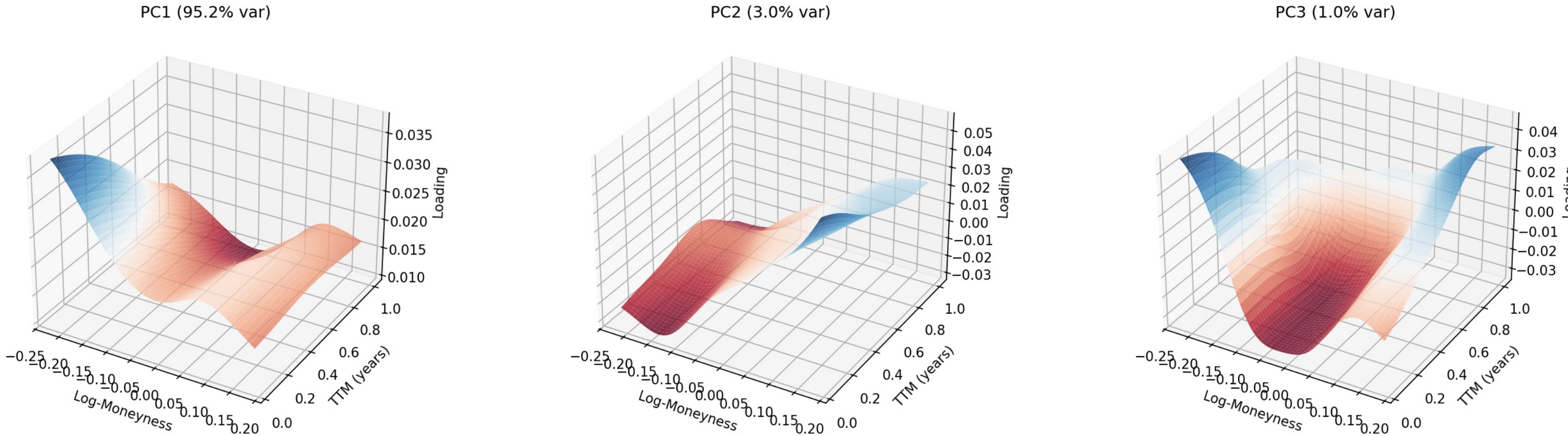
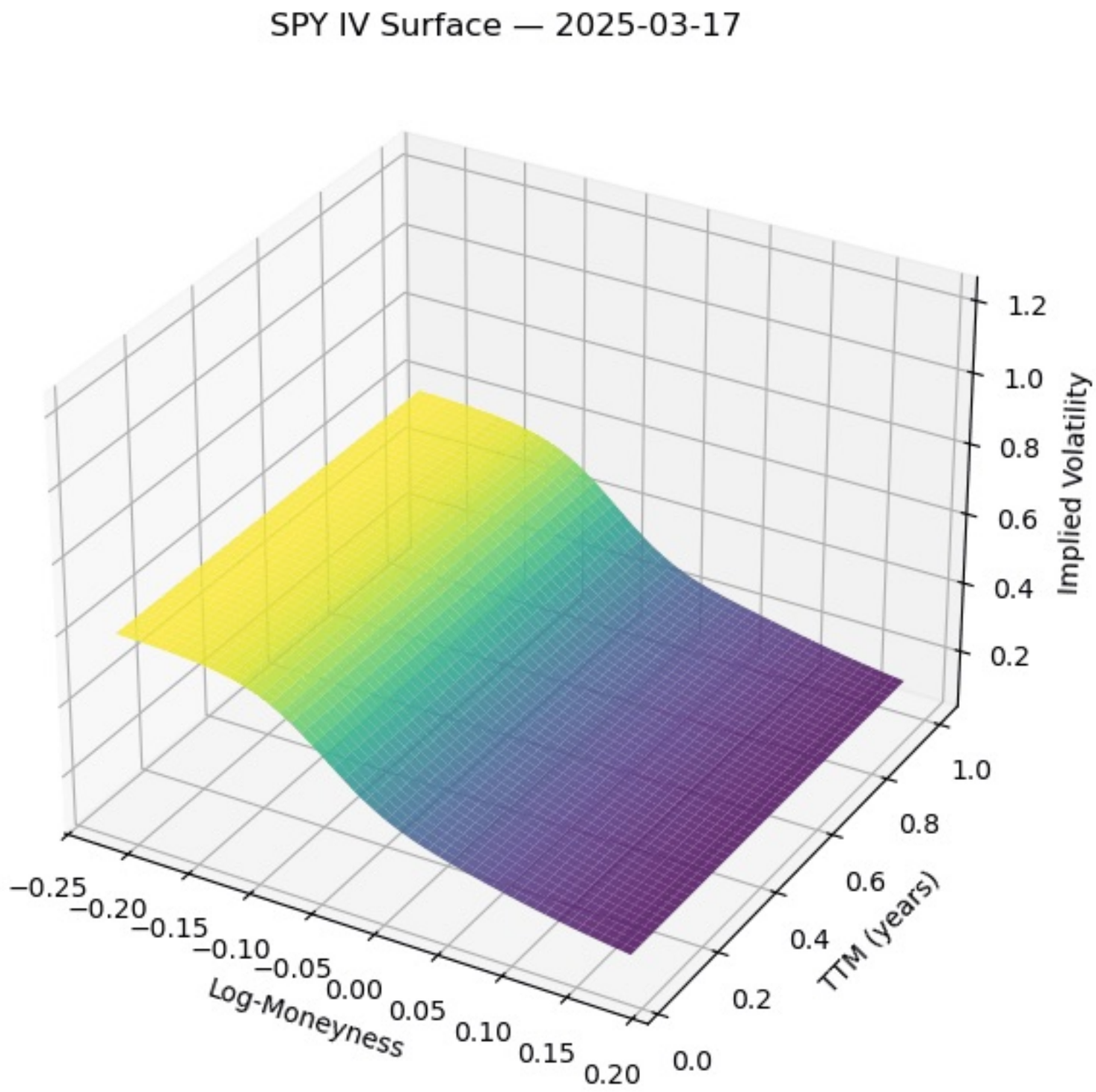
Component	Variance Explained	Cumulative
PC1 (Level)	~95%	~95%
PC2 (Slope)	~3%	~98%
PC3 (Curvature)	~1%	~99%


Mean SPY IV Surface — 2025
IV
0.28
0.26
0.24
0.22
0.2
0.18
0.16
0.14
Implied Volatility
0.25
0.2
0.15
−0.2
0
Log-Moneyness (log(K/S))
0.2
0.4
0.6
0.8
1
Time to Maturity (years)

Principal Component Loadings — SPY IV Surface 2025




SPY IV Surface — 2025-03-17
Implied Volatility
1.2
1.0
0.8
0.6
0.4
0.2
TTM (years)
0.0
0.2
0.4
0.6
0.8
1.0
Log-Moneyness
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20

Principal Component Scores — SPY IV Surface 2025
PC1 Score Time Series (95.2% variance explained)
PC1 Score
25
20
15
10
5
0
−5
PC2 Score Time Series (3.0% variance explained)
PC2 Score
1
0
−1
−2
−3
−4
−5
PC3 Score Time Series (1.0% variance explained)
PC3 Score
1.5
1.0
0.5
0.0
−0.5
−1.0
2025-04
2025-05
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
Date